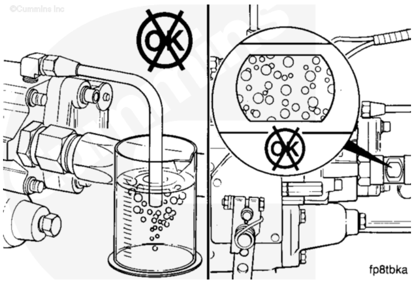
Trainings

General Information

There are two methods to check for air in the fuel suction line.

- Gear pump drain method
- Sight glass method.

Both methods are described in this procedure.

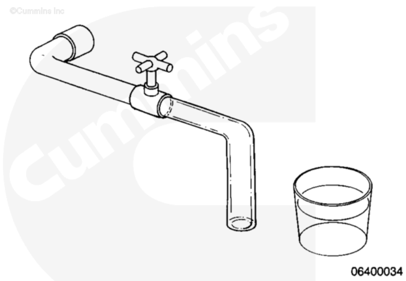


Leak Test

Gear Pump Drain Method

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to avoid severe personal injury or death when working on the fuel system.

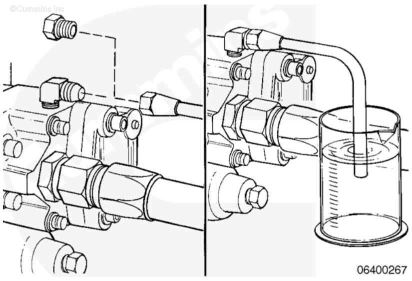


To perform a pressure-side air-in-fuel test, use the following items:

- Quick-disconnect fitting, Part Number 3376859
- High-pressure hose
- Pressure valve (Capable of 2758 kPa [400 psi])
- Clear tubing
- Clean container.

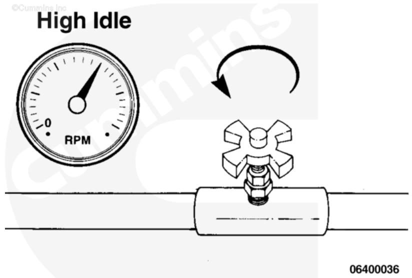
Connect the equipment to the quick-connect fitting at the fuel pump outlet.

Put the end of the clear hose in the clean container.



Operate the engine at high idle with no load.

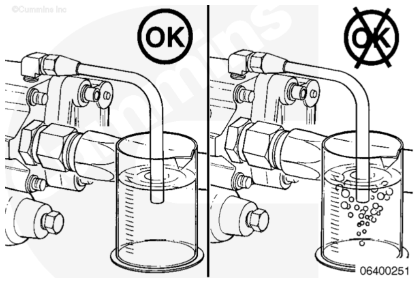
Slowly open the valve until a steady stream of fuel is visible.



Verify the hose is below the surface of the fuel in the container.

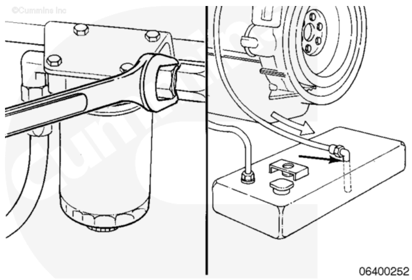
If air is in the fuel pump suction line, bubbles will be visible in the fuel.

Note : If the application incorporates a Day tank or a Positive Head Fuel System it is recommended to use the Sight Glass Method placed before this accessory as these devices may mask the presence of air in fuel upstream.



If an air leak is found perform the following:

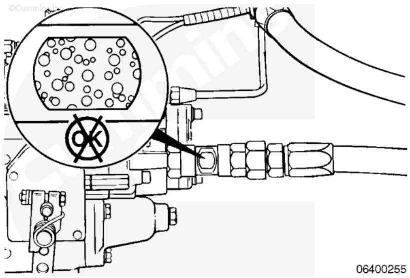
- Systematically inspect the entire fuel supply routing for sources of air ingress starting at the fuel tank followed by all hose/tube interconnections, fuel filtration hardware and day tank/positive head fuel system, if provisioned. Tighten any loose connections as needed.
- Inspect drop tube in the fuel tank for damage.
- Inspect fuel return to tank and verify the tube is both above fuel level and at a minimum distance of 305 mm [12 in] from the fuel supply connection.
- Inspect the o-rings for damage.
- Correct the air leak and test for other leaks.
- Remove the test equipment.
- Retest engine duplicating the operating conditions when performance complaint occurred to confirm air in fuel correction. This is especially important for standby



Generator applications with infrequent starts and may require testing after an extended rest period.

Sight Glass Method

Remove the fuel inlet line.



Install sight glass, Part Number 3163270, at the inlet of the fuel pump.

Place a source a light on the opposite side from the side viewed of the sight glass.

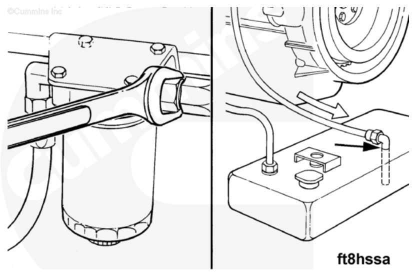
Operate the engine at high idle with no load, or at the full fuel position.

A small air leak will have a milky appearance.

A large air leak will look like bubbles in the fuel.

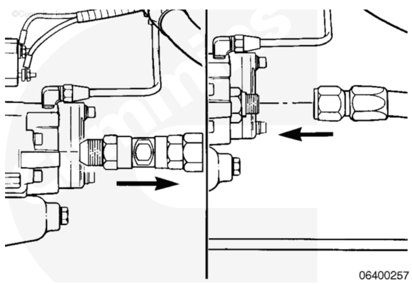
Note : If the application incorporates a Day tank or a Positive Head Fuel System it is recommended to apply a sight glass before this accessory as these devices may mask the presence of air in fuel upstream.

If an air leak is found, perform the following:



- Systematically inspect the entire fuel supply routing for sources of air ingress starting at the fuel tank followed by all hose/tube interconnections, fuel filtration hardware and day tank/positive head fuel system, if provisioned. Tighten any loose connections as needed.
- Inspect drop tube in the fuel tank for damage.
- Inspect fuel return to tank and ensure the tube is both above fuel level and at a minimum distance of 305 mm [12 in] from the fuel supply connection.
- Inspect o-rings for damage.

Continue to test and look for source of air until no air bubbles are visible.



Remove the sight glass and install the inlet hose.

Tighten the fuel inlet hose.

Torque Value: 120 n•m [89 ft-lb]

Retest engine duplicating the operating conditions when performance complaint occurred to confirm air in fuel correction. This is especially important for standby Generator

applications with infrequent starts and may require testing
after an extended rest period.

Last Modified: 02-Jan-2023